

HMI (Human–Machine Interface)

Workshop Notes for Enthusiasts

1. Popular HMI Manufacturers & Their Software

Different automation companies provide their own HMI hardware along with dedicated configuration software:

1. Mitsubishi Electric
 - HMI: *GOT (Graphic Operation Terminal)*
 - Software: *GT Designer*
2. Allen-Bradley (Rockwell Automation)
 - HMI: *PanelView Series*
 - Software: *PanelBuilder / FactoryTalk View*
3. Siemens
 - HMI: *OP/TP Series (Operator Panels)*
 - Software: *WinCC Flexible / TIA Portal*
4. Beijer Electronics
 - HMI: *E-Terminal*
 - Software: *E-Designer*
5. Delta Electronics
 - HMI: *DOP Series*
 - Software: *DOPSoft*

2. PLC–HMI Interfacing (Important Concept)

One of the biggest advantages of modern HMIs is flexibility:

-  Any brand of PLC can be interfaced with any brand of HMI
- The main requirement is:

👉 The HMI software must support the PLC driver for that PLC brand/model.

What is a PLC Driver?

- A PLC driver is a communication protocol file that allows the HMI to read and write data from the PLC.
- Examples:
 - Modbus (RTU/TCP)
 - Siemens S7 Protocol
 - Mitsubishi MC Protocol
 - Allen-Bradley Ethernet/IP

🔴 If the driver is available in the HMI software, communication is possible—regardless of brand.

3. Why Do We Use an HMI?

An HMI acts as the bridge between humans and machines.

1. Monitoring Process Data
 - Display real-time values such as temperature, pressure, speed, levels, and alarms.
 2. Controlling the Process
 - Change numerical values (setpoints, timers, counters).
 - Start/stop machines and reset faults.
 3. Data Scaling & Representation
 - Convert raw PLC values into meaningful units.
 - Example:
 - PLC value 0–10000 → HMI display 0.0–100.0 °C
 - Scaling range can be $0.1\times$ up to $100,000\times$ or more, depending on application needs.
 4. Interfacing with PLC I/O
 - Identify PLC input and output terminals using tags or addresses.
 - Helps in commissioning, testing, and troubleshooting.
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4. Key Takeaway for Enthusiasts

- PLC = Brain 🧠
- Sensors & Actuators = Muscles 💪
- HMI = Control Dashboard 👁️ 🖨️

A well-designed HMI:

- Makes machines easier to operate
- Reduces downtime
- Improves safety
- Allows multi-brand system integration